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Yamashita et al.

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(54) **OPIOID RECEPTOR LIGANDS AND METHODS OF USING AND MAKING SAME**

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Related U.S. Application Data

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A61K 45/06 (2006.01)

A61K 31/444 (2006.01)

A61K 31/4433 (2006.01)

A61K 31/35 (2006.01)

A61K 9/00 (2006.01)

C07D 409/12 (2006.01)

C07D 409/14 (2006.01)

C07D 405/14 (2006.01)

C07D 405/12 (2006.01)

C07D 405/04 (2006.01)

C07D 311/96 (2006.01)

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CPC **A61K 31/4436** (2013.01); **A61K 9/0019** (2013.01); **A61K 31/35** (2013.01); **A61K 31/444** (2013.01); **A61K 31/4433** (2013.01);

A61K 45/06 (2013.01); **C07D 311/96** (2013.01); **C07D 405/04** (2013.01); **C07D 405/12** (2013.01); **C07D 405/14** (2013.01); **C07D 409/12** (2013.01); **C07D 409/14** (2013.01)

(58) **Field of Classification Search**

CPC .. **A61K 31/4436**; **A61K 9/0019**; **A61K 31/35**; **A61K 31/4433**; **A61K 31/444**; **A61K 45/06**; **C07D 311/96**; **C07D 405/04**; **C07D 405/12**; **C07D 405/14**; **C07D 409/12**; **C07D 409/14**
USPC 514/161
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,115,538 A	9/1978	Satoh et al.
5,436,128 A	7/1995	Harpold et al.
6,071,918 A	6/2000	Cook
6,136,839 A	10/2000	Isakson et al.
7,645,767 B2	1/2010	Singh et al.
8,835,488 B2	9/2014	Yamashita et al.
9,044,469 B2	6/2015	Yamashita et al.
9,309,234 B2	4/2016	Yamashita et al.
9,642,842 B2	5/2017	Yamashita et al.
9,849,119 B2	12/2017	Yamashita et al.
2006/0217372 A1	9/2006	Blanco-Pillado et al.
2009/0028873 A1	1/2009	Gant et al.
2009/0247561 A1	10/2009	Zemolka et al.
2010/0280058 A1	11/2010	Dolle et al.
2013/0158033 A1	6/2013	Vela Hernandez et al.
2015/0246904 A1	9/2015	Yamashita et al.

FOREIGN PATENT DOCUMENTS

EP	1562595 A1	8/2005
JP	S60139684 A	7/1985
JP	2005514389 A	5/2005
JP	2006511474 A	4/2006
WO	1996005166 A1	2/1996
WO	2003063797 A2	8/2003
WO	2007041341 A2	4/2007
WO	2007081966 A2	7/2007
WO	2008057856 A2	5/2008
WO	2008057857 A1	5/2008
WO	2009118173 A1	10/2009

(Continued)

OTHER PUBLICATIONS

APOLLO-1 topline public announcement, Trevena Inc., Feb. 21, 2017.

Arutyunyan, N.S. et al., "Syntheses and Some Transformations of 4-Aryl-Substituted Amines of the Tetrahydropyran Series", Russian Journal of organic Chemistry, vol. 47 No. 1 (2011) pp. 115-119.

(Continued)

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(57) **ABSTRACT**

This application describes compounds that can act as opioid receptor ligands, which compounds can be used in the treatment of, for example, pain and pain related disorders.

17 Claims, No Drawings

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S—O; S—S; S—N fragments in the ring structure are excluded from this composition.

The ring containing A₁, A₂, A₃, A₄, A₅ and the carbon connected to D₁ can be fused with another ring, such as benzene, pyridine, pyrimidine, furan, thiophene or pyridazine, but not limited to these examples, where the resulting bicycle is chemically stable and synthetically accessible. It is also understood that the above-mentioned fused rings could be multiply substituted with cyano, halogen, alkyl, branched alkyl, halogenated alkyl, hydroxyl, alkoxy, formyl, acetyl, amino, alkylamino, dialkylamino, mercaptanyl, alkylmercaptanyl, and other small substitution groups. The bonds between A₁ and A₂, A₂ and A₃, A₃ and A₄, A₄ and A₅ can independently be a single bond or a double bond. The bonds between A₁ and A₂, A₂ and A₃, A₃ and A₄, A₄ and A₅ cannot be a double bond at the same time.

A₂ and A₄ can be connected by a carbon bridge. Examples of such a bridge include —CH₂—, and —CH₂CH₂—.

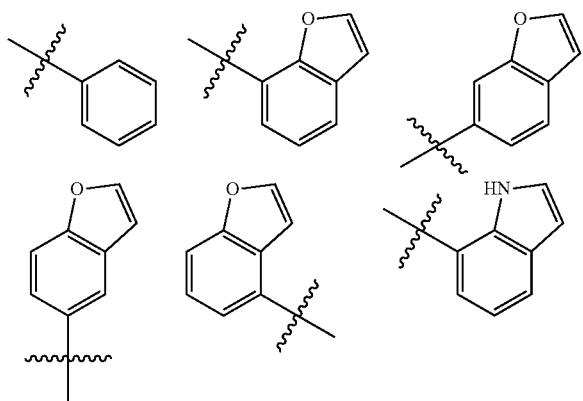
B₁ is CH₂, CHR₁₃, CR₁₃R₁₄, O, S, SO, SO₂, NH, NR₁₃, CR₁₃ or CO. B₂ is CH₂, CHR₁₅, CR₁₅R₁₆, CR₁₅ or CO. B₃ is H, alkyl, branched alkyl, halogenated alkyl, aryl, arylalkyl, alkylcarbonyl, branched alkylcarbonyl, arylcarbonyl, alkoxy carbonyl, or alkylsulfonyl. B₄ is null, C₁–C₆ alkyl, CH₂, CH₂CH₂, CHR₁₉, CR₁₉R₂₀ or CO. In some embodiments, when B₄ is an alkyl one or more of the hydrogens can be replaced with a deuterium. B₅ is alkyl, branched alkyl, halogenated alkyl, carbocycle-substituted alkyl, aryl, carbocycle or arylalkyl.

Aryl, carbocycle (non-aromatic)/heterocycle (non-aromatic with 1-3 heteroatoms, including O, N, S) are either unsubstituted, or substituted with small substitution groups. Small substitution groups can be cyano, halogen, alkyl, branched alkyl, halogenated alkyl, hydroxyl, alkoxy, amino, alkylamino, dialkylamino, mercaptanyl, alkylmercaptanyl, alkylsulfonyl, aminosulfonyl, alkylaminosulfonyl, alkylcarbonyl, alkoxy carbonyl, aminocarbonyl, alkylaminocarbonyl, dialkylaminocarbonyl, aryl, arylalkyl, carbocycle or carbocycle-alkyl. In some embodiments, the small substitution groups are selected from F, Cl, Br, CH₃, CH₂CH₃, CH₂F, CHF₂, CF₃, n-Pr, n-Bu, i-Bu, sec-Bu, t-Bu, CN, OH, OMe, OEt, O-iPr, OCF₃, NH₂, NHMe, NMe₂, methoxycarbonyl, methanesulfonyl, Ph, benzyl, MeSO₂, formyl, and acetyl.

Carbocycle may contain double bonds, but they should not be aromatic.

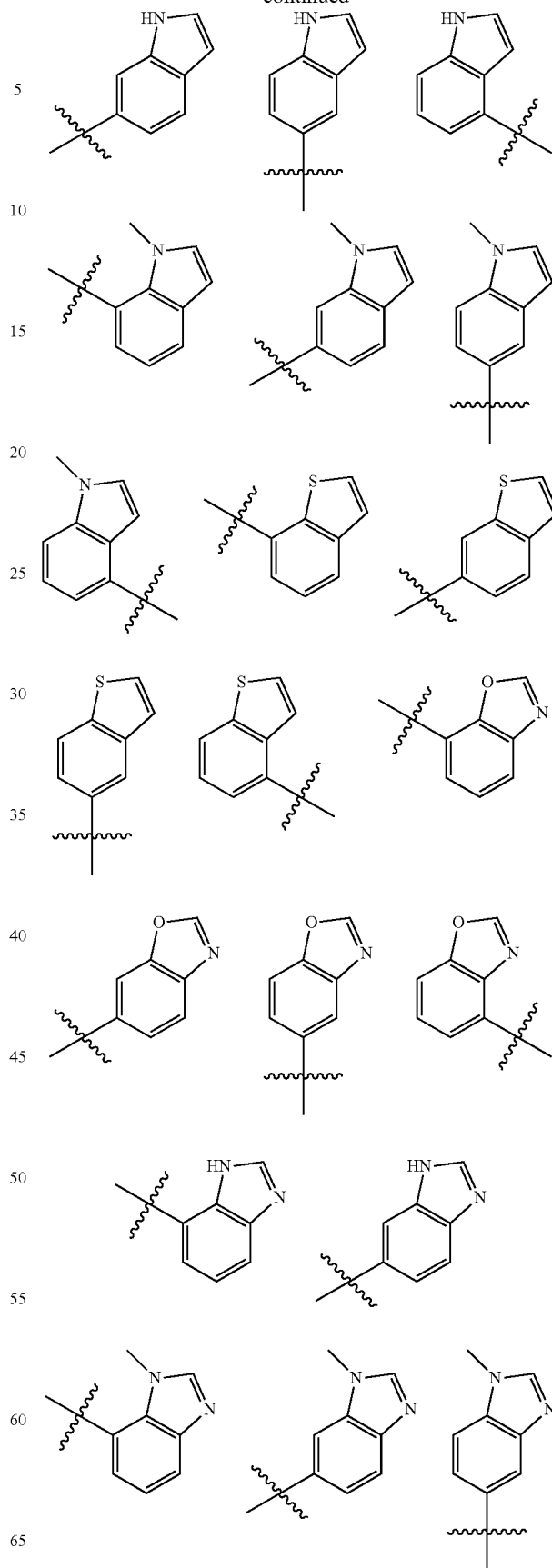
D₁ is an aryl group or a carbocycle.

An aryl group is either a monocyclic aromatic group or a bicyclic aromatic group, which may contain heteroatoms in the aromatic group (e.g. heteroaryl). The following structures are some examples of representative aryl groups, but the aryl groups are not limited to those examples:

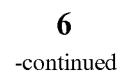


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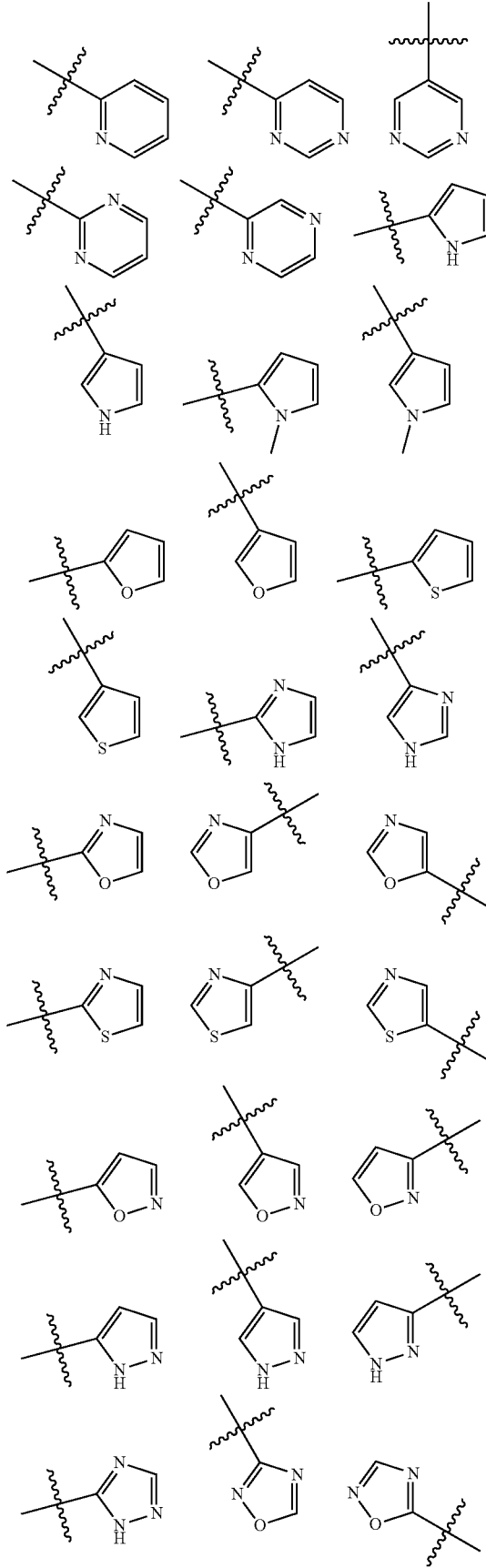
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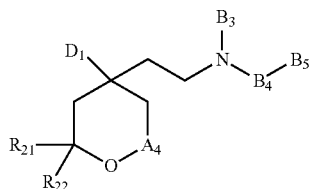


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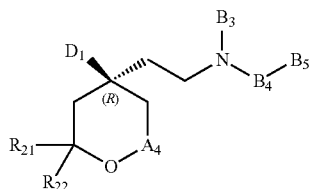
Further D₁ is an aryl. Examples of the aryl groups are shown above.

The aryl groups can be mono or multiply substituted with F, Cl, Br, CH₃, CH₂CH₃, CH₂F, CHF₂, CF₃, n-Pr, n-Bu, i-Bu, sec-Bu, i-Pr, t-Bu, CN, OH, OMe, OEt, O-iPr, OCF₃, NH₂, NHMe, NMe₂, methoxycarbonyl, Ph, benzyl, formyl, or acetyl.

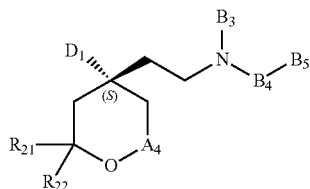
This application also describes compounds having the structure of Formula IV-1, IV-2, or IV-3, V, or VI:



IV-1



IV-2



IV-3

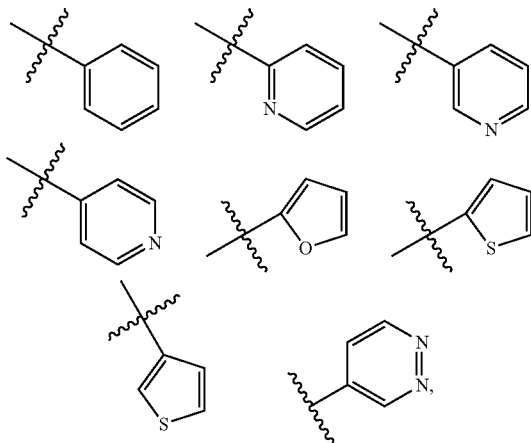
wherein R₂₁ and R₂₂ are, independently, H or CH₃; A₄ is CH₂, CR₉R₁₀ or a cycle of the formula C(CH₂)_n, where n=2-5.

Further R₉ and R₁₀ are independently CH₃ or CH₂CH₃.

Further B₃ is H, C₁-C₆ alkyl or branched alkyl.

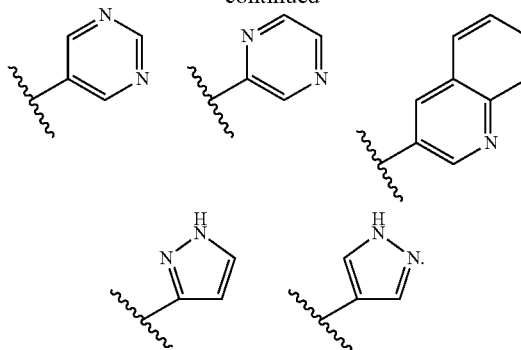
Further B₄ is null, C₁-C₆ alkyl, CH₂, CH₂CH₂, or —CHCH₃.

B₅ is —(CH₂)_nCH₃, where n=2-3, —C(CH₃)₃, cyclohexyl, cyclopentyl, aryl or arylalkyl. The aryl group can be selected from the list below:



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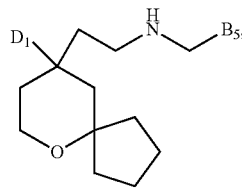
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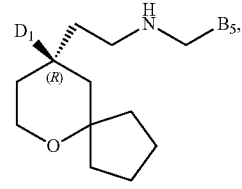
Each aryl groups can be mono or multiply substituted with F, I, Cl, Br, CH₃, CN, OH, OMe, OEt, OCF₃, CF₃, or methanesulfonyl.

Further, in some embodiments, D₁ is a phenyl, 2-pyridyl, 3-pyridyl, or 4-pyridyl which can be independently mono or multiply substituted with F, Cl, Br, OCF₃, CF₃, or CH₃.

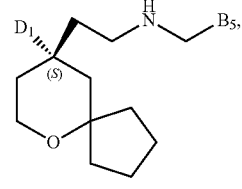
This application also describes compounds having the structure of Formula V-1, V-2, V-3, VI-1, VI-2, or VI-3:



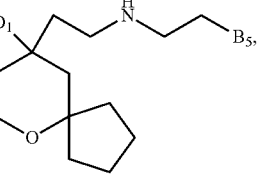
V-1



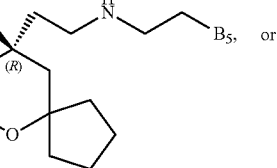
V-2



V-3



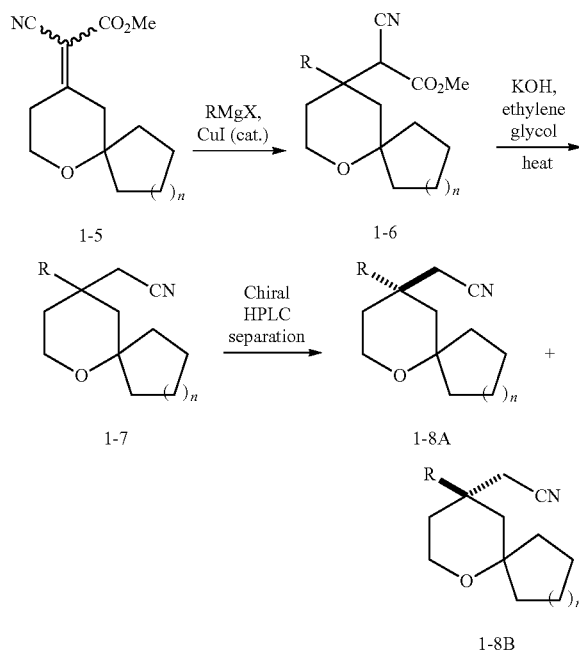
VI-1



VI-2

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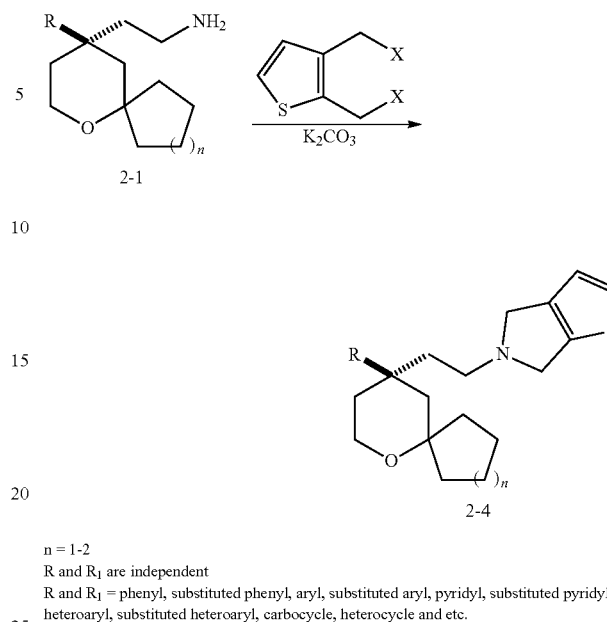
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 $n = 1-2$

R = phenyl, substituted phenyl, aryl, substituted aryl, pyridyl, substituted pyridyl, heteroaryl, substituted heteroaryl, carbocycle, heterocycle and etc.

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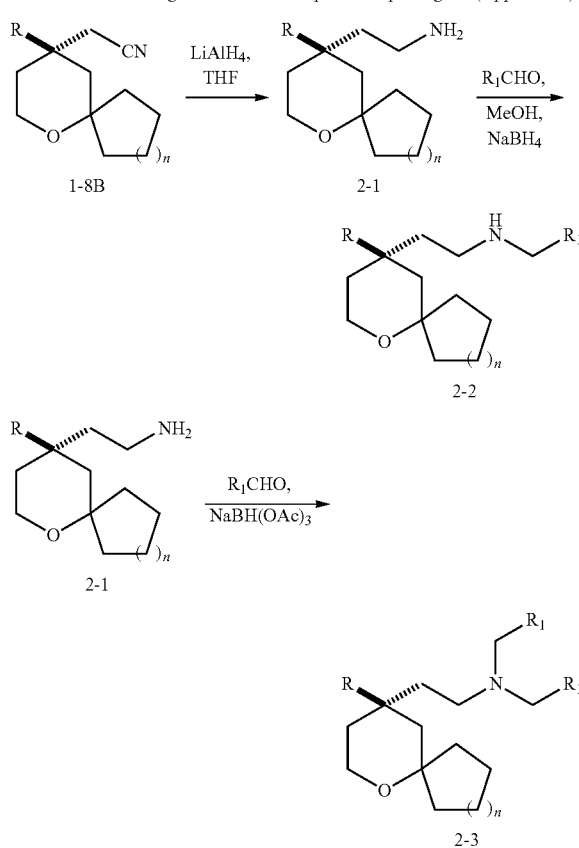


In some embodiments, the same scheme is applied to 1-7 and 1-8A.

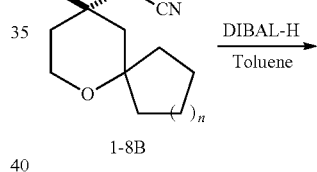
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Scheme 3: Converting the nitrile to the opioid receptor ligand (Approach 2)

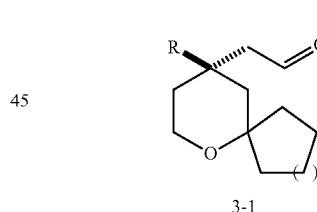
Scheme 2: Converting the nitrile to the opioid receptor ligand (Approach 1)



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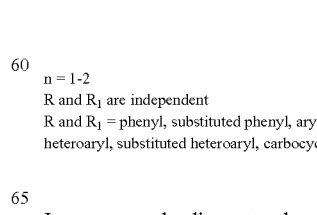
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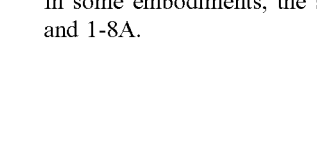
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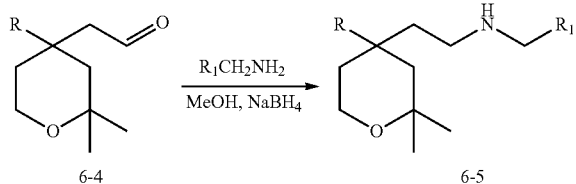


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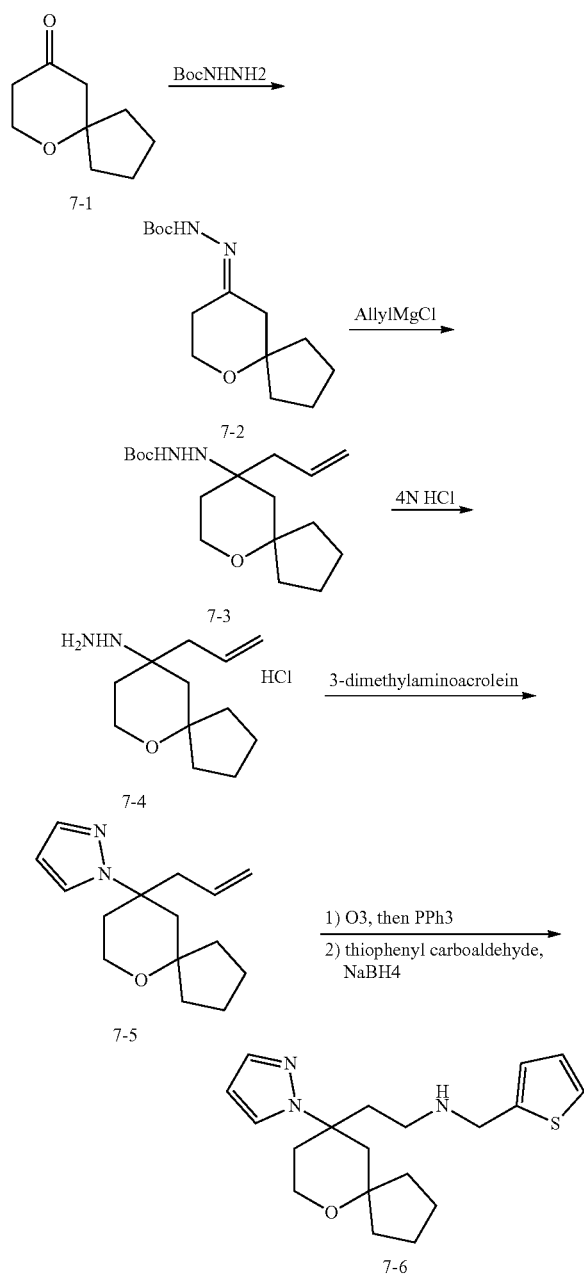
In some embodiments, the same scheme is applied to 1-7 and 1-8A.

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 $n = 1-2$ R and R₁ are independentR and R₁ = phenyl, substituted phenyl, aryl, substituted aryl, pyridyl, substituted pyridyl, heteroaryl, substituted heteroaryl, carbocycle, heterocycle and etc.

Scheme 7: N-linked Pyrazole Opioid Receptor Ligand



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Scheme 8

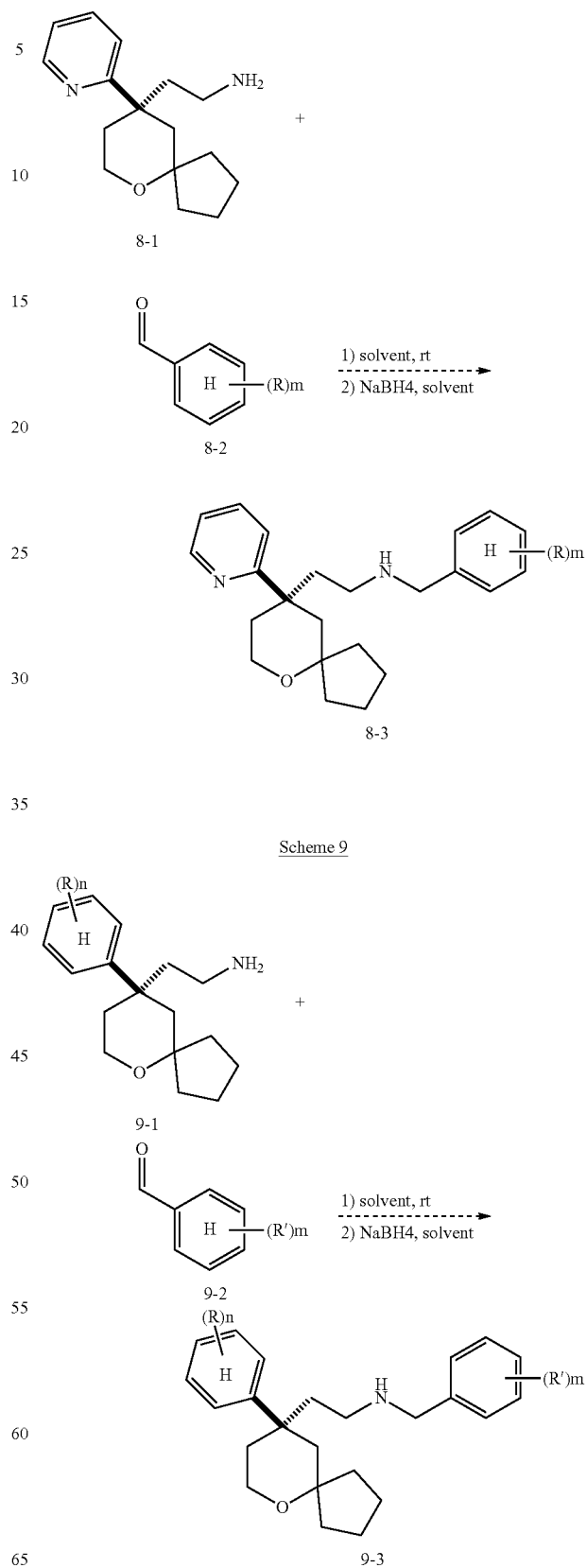


TABLE 2-continued

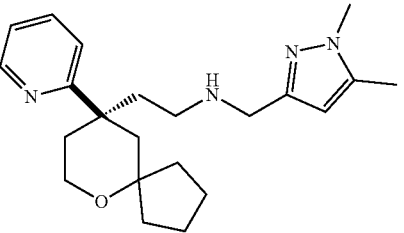
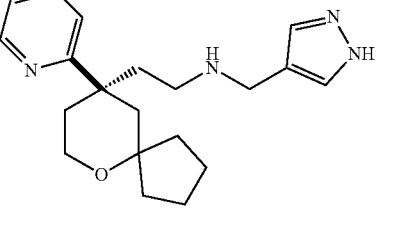
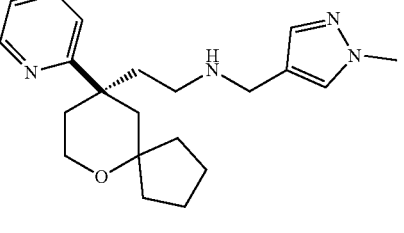
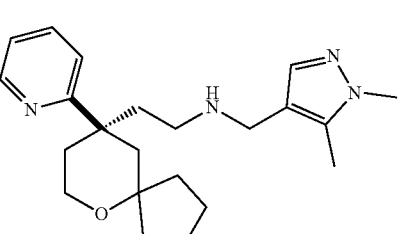
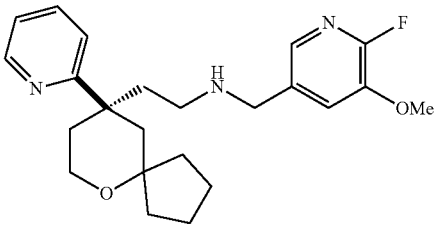
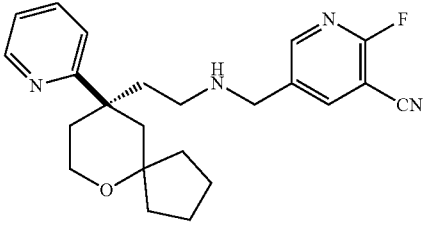
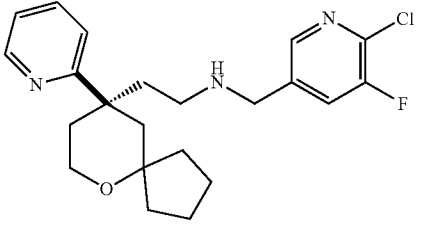
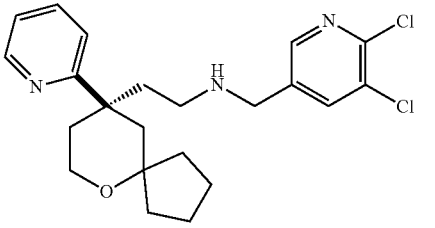
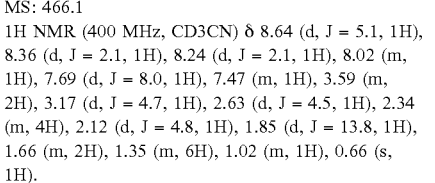
Compound Name	Structure and/or NMR Spectrum
526. (1H-pyrazol-4-ylmethyl)({2-[(9R)-9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]ethyl})amine	 MS: 341.2 ¹H NMR (400 MHz, CD₃CN) δ 8.73 (dd, J = 5.3, 1.1, 1H), 8.21 (td, J = 8.0, 1.7, 2H), 7.75 (d, J = 8.2, 1H), 7.66 (m, 3H), 7.56 (s, 1H), 3.96 (s, 2H), 3.73 (m, 2H), 2.91 (m, 1H), 2.32 (m, 4H), 2.08 (m, 1H), 1.99 (m, 1H), 1.78 (m, 2H), 1.62 (m, 1H), 1.49 (m, 4H), 1.19 (m, 1H), 0.78 (dt, J = 13.1, 8.8, 1H).
527. [(1-methyl-1H-pyrazol-4-yl)methyl]({2-[(9R)-9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]ethyl})amine	 MS: 355.2 ¹H NMR (400 MHz, CD₃CN) δ 8.78 (dd, J = 5.5, 1.3, 1H), 8.32 (td, J = 8.0, 1.7, 1H), 7.84 (d, J = 8.2, 1H), 7.77 (ddd, J = 7.6, 5.5, 0.9, 1H), 7.71 (brs, 1H), 7.55 (s, 1H), 7.43 (s, 1H), 3.91 (s, 2H), 3.81 (d, J = 11.7, 3H), 3.73 (m, 2H), 2.90 (dt, J = 11.7, 5.8, 1H), 2.35 (m, 4H), 2.14 (ddd, J = 10.8, 10.2, 5.2, 1H), 2.03 (d, J = 14.2, 1H), 1.80 (m, 2H), 1.62 (tdd, J = 8.7, 6.8, 2.7, 1H), 1.49 (m, 4H), 1.20 (m, 1H), 0.80 (dt, J = 12.9, 8.8, 1H).
528. [(5-methyl-1H-pyrazol-4-yl)methyl]({2-[(9R)-9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]ethyl})amine	
529. [(1,5-dimethyl-1H-pyrazol-4-yl)methyl]({2-[(9R)-9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]ethyl})amine	

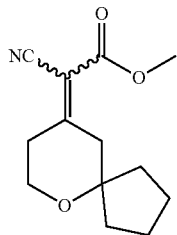
TABLE 2-continued

Examples with chemical name and/or characterization data	
Compound Name	Structure and/or NMR Spectrum
535. [(6-fluoro-5-methoxypyridin-3-yl)methyl]({2-[(9R)-9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]ethyl})amine	
536. 2-fluoro-5-[(2-[(9R)-9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]ethyl)amino)methyl]pyridine-3-carbonitrile	
537. [(6-chloro-5-fluoropyridin-3-yl)methyl]({2-[(9R)-9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]ethyl})amine	MS: 404.2 ¹H NMR (400 MHz, CD₃CN) δ 8.68 (dd, J = 5.2, 1.1, 1H), 8.24 (d, J = 1.9, 1H), 8.10 (td, J = 7.9, 1.8, 1H), 7.78 (dd, J = 9.0, 2.0, 1H), 7.69 (d, J = 8.2, 1H), 7.56 (ddd, J = 7.5, 5.3, 0.9, 1H), 4.75 (brs, 1H), 4.07 (m, 2H), 3.72 (m, 2H), 2.99 (td, J = 11.9, 5.2, 1H), 2.48 (td, J = 12.0, 4.5, 1H), 2.32 (m, 3H), 2.11 (m, 1H), 1.77 (m, 2H), 1.62 (m, 1H), 1.49 (m, 4H), 1.17 (m, 1H), 0.77 (dt, J = 13.1, 8.9, 1H). 
538. [(5,6-dichloropyridin-3-yl)methyl]({2-[(9R)-9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]ethyl})amine	MS: 422.2 ¹H NMR (400 MHz, CD₃CN) δ 8.51 (m, 1H), 8.33 (d, J = 2.1, 1H), 8.12 (d, J = 2.1, 1H), 7.76 (t, J = 7.9, 1H), 7.49 (d, J = 8.1, 1H), 7.23 (dd, J = 7.4, 4.9, 1H), 4.26 (d, J = 1.5, 2H), 3.57 (dd, J = 7.7, 3.0, 2H), 3.09 (td, J = 12.2, 4.6, 1H), 2.55 (td, J = 12.1, 4.6, 1H), 2.27 (dddd, J = 25.5, 17.3, 14.3, 3.4, 4H), 1.77 (m, 1H), 1.59 (m, 2H), 1.34 (m, 6H), 0.98 (dd, J = 11.4, 5.0, 1H), 0.60 (dt, J = 13.4, 9.0, 1H). 
539. [(5-bromo-6-chloropyridin-3-yl)methyl]({2-[(9R)-9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]ethyl})amine	MS: 466.1 ¹H NMR (400 MHz, CD₃CN) δ 8.64 (d, J = 5.1, 1H), 8.36 (d, J = 2.1, 1H), 8.24 (d, J = 2.1, 1H), 8.02 (m, 1H), 7.69 (d, J = 8.0, 1H), 7.47 (m, 1H), 3.59 (m, 2H), 3.17 (d, J = 4.7, 1H), 2.63 (d, J = 4.5, 1H), 2.34 (m, 4H), 2.12 (d, J = 4.8, 1H), 1.85 (d, J = 13.8, 1H), 1.66 (m, 2H), 1.35 (m, 6H), 1.02 (m, 1H), 0.66 (s, 1H). 

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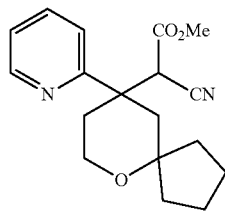
Example 15: Synthesis of [(3-methoxythiophen-2-yl)methyl]({2-[(9R)-9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]ethyl})amine (Compound 140)

Methyl 2-cyano-2-[6-oxaspiro[4.5]decan-9-ylidene]acetate (Mixture of E and Z Isomers)



A mixture of 6-oxaspiro[4.5]decan-9-one (13.74 g, 89.1 mmol), methylcyanoacetate (9.4 ml, 106.9 mmol), ammonium acetate (1.79 g, 26.17 mmol) and acetic acid (1.02 ml, 17.8 mmol) in benzene (75 ml) was heated at reflux in a 250 ml round bottom flask equipped with a Dean-Stark and a reflux condenser. After 3 h, TLC (25% EtOAc in hexane, PMA stain) showed the reaction was completed. After cooling, benzene (50 ml) was added and the layer was separated, the organic was washed by water (120 ml) and the aqueous layer was extracted by CH_2Cl_2 (3x120 ml). The combined organic was washed with sat'd NaHCO_3 , brine, dried and concentrated and the residual was purified by flash chromatography (340 g silica gel column, eluted by EtOAc in hexane: 5% EtOAc, 2CV; 5-25%, 14CV; 25-40%, 8 CV) gave a mixture of E and Z isomers: methyl 2-cyano-2-[6-oxaspiro[4.5]decan-9-ylidene]acetate (18.37 g, 87.8% yield, m/z 236.0 $[\text{M}+\text{H}]^+$ observed) as a clear oil.

Methyl 2-cyano-2-[9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]acetate

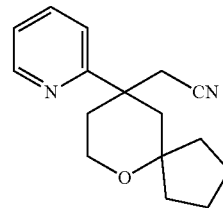


A solution of 2-bromopyridine (14.4 ml, 150 mmol) in THF (75 ml) was added dropwise to a solution of isopropylmagnesium chloride (75 ml, 2M in THF) at 0° C. under N_2 , the mixture was then stirred at rt for 3 h, copper iodide (2.59 g, 13.6 mmol) was added and allowed to stir at rt for another 30 min before a solution of a mixture of E and Z isomers of methyl 2-cyano-2-[6-oxaspiro[4.5]decan-9-ylidene]acetate (16 g, 150 mmol) in THF (60 ml) was added in 30 min. The mixture was then stirred at rt for 18 h. The reaction mixture was poured into a 200 g ice/2 N HCl (100 ml) mixture. The product was extracted with Et_2O (3x300 ml), washed with brine (200 ml), dried (Na_2SO_4) and

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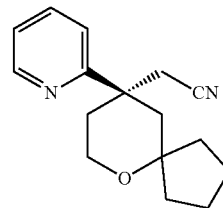
concentrated. The residual was purified by flash chromatography (100 g silica gel column, eluted by EtOAc in hexane: 3% 2CV; 3-25%, 12 CV; 25-40% 6CV gave methyl 2-cyano-2-[9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]acetate (15.44 g, 72% yield, m/z 315.0 $[\text{M}+\text{H}]^+$ observed) as an amber oil.

2-[9-(Pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]acetonitrile



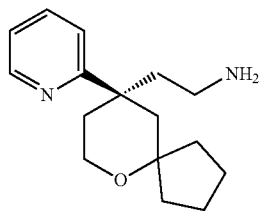
Ethylene glycol (300 ml) was added to methyl 2-cyano-2-[9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]acetate (15.43 g, 49 mmol) followed by potassium hydroxide (5.5 g, 98 mmol), the resulting mix was heated to 120° C., after 3 h, the reaction mix was cooled and water (300 ml) was added, the product was extracted by Et_2O (3x400 ml), washed with water (200 ml), dried (Na_2SO_4) and concentrated, the residual was purified by flash chromatography (340 g silica gel column, eluted by EtOAc in hexane: 3% 2CV; 3-25%, 12 CV; 25-40% 6CV to give 2-[9-(Pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]acetonitrile (10.37 g, 82% yield, m/z 257.0 $[\text{M}+\text{H}]^+$ observed).

2-[(9R)-9-(Pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]acetonitrile



The racemic 2-[9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]acetonitrile was separated by chiral HPLC column under the following preparative-SFC conditions: Instrument: SFC-80 (Thar, Waters); Column: Chiralpak AD-H (Daicel); column temperature: 40° C.; Mobile phase: Methanol/ CO_2 =40/60; Flow: 70 g/min; Back pressure: 120 Bar; Cycle time of stack injection: 6.0 min; Load per injection: 225 mg; Under these conditions, 2-[9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]acetonitrile (4.0 g) was separated to provide the desired isomer, 2-[(9R)-9-(Pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]acetonitrile (2.0 g, >99.5% enantiomeric excess) as a slow-moving fraction. The absolute (R) configuration of the desired isomer was later determined by an X-ray crystal structure analysis of Compound 140.

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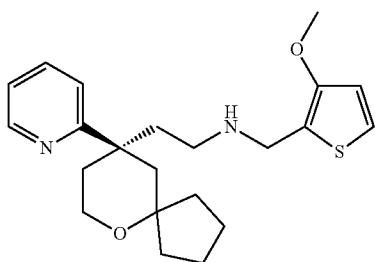
2-[(9R)-9-(Pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]
ethan-1-amine

LAH (1M in Et₂O, 20 ml, 20 mmol) was added to a solution of 2-[(9R)-9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]acetonitrile (2.56 g, 10 mmol) in Et₂O (100 ml, 0.1M) at 0° C. under N₂. The resulting mix was stirred and allowed to warm to room temperature. After 2 h, LCMS showed the reaction had completed. The reaction was cooled at 0° C. and quenched with water (1.12 ml), NaOH (10%, 2.24 ml) and another 3.36 ml of water. Solid was filtered and filter pad was washed with ether (3×20 ml). The combined organic was dried and concentrated to give 2-[(9R)-9-(Pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]ethan-1-amine (2.44 g, 94% yield, m/z 260.6 [M+H]⁺ observed) as a light amber oil.

Alternatively, 2-[(9R)-9-(Pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]ethan-1-amine was prepared by Raney-Nickel catalyzed hydrogenation.

An autoclave vessel was charged with 2-[(9R)-9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl] acetonitrile and ammonia (7N solution in methanol). The resulting solution was stirred at ambient conditions for 15 minutes and treated with Raney 2800 Nickel, slurried in water. The vessel was pressurized to 30 psi with nitrogen and agitated briefly. The autoclave was vented and the nitrogen purge repeated additional two times. The vessel was pressurized to 30 psi with hydrogen and agitated briefly. The vessel was vented and purged with hydrogen two additional times. The vessel was pressurized to 85-90 psi with hydrogen and the mixture was warmed to 25-35° C. The internal temperature was increased to 45-50° C. over 30-60 minutes. The reaction mixture was stirred at 45-50° C. for 3 days. The reaction was monitored by HPLC. Once reaction was deemed complete, it was cooled to ambient temperature and filtered through celite. The filter cake was washed with methanol (2×). The combined filtrates were concentrated under reduced pressure at 40-45° C. The resulting residue was co-evaporated with EtOH (3×) and dried to a thick syrupy of 2-[(9R)-9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]ethan-1-amine.

[(3-Methoxythiophen-2-yl)methyl]({2-[(9R)-9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]ethyl})amine



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Into a vial were added 2-[(9R)-9-(Pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]ethan-1-amine (500 mg, 1.92 mmole), 18 mL CH₂Cl₂ and sodium sulfate (1.3 g, 9.6 mmole). The 3-methoxythiophene-2-carboxaldehyde (354 mg, 2.4 mmole) was then added, and the mixture was stirred overnight. NaBH₄ (94 mg, 2.4 mmole) was added to the reaction mixture, stirred for 10 minutes, and then MeOH (6.0 mL) was added, stirred 1 h, and finally quenched with water. The organics were separated off and evaporated. The crude residue was purified by a Gilson prep HPLC. The desired fractions collected and concentrated and lyophilized. After lyophilization, residue was partitioned between CH₂Cl₂ and 2N NaOH, and the organic layers were collected. After solvent was concentrated to half of the volume, 1.0 eq of 1N HCl in Et₂O was added, and majority of solvent evaporated under reduced pressure. The solid obtained was washed several times with Et₂O and dried to provide [(3-methoxythiophen-2-yl)methyl]({2-[(9R)-9-(pyridin-2-yl)-6-oxaspiro[4.5]decan-9-yl]ethyl})amine monohydrochloride (336 mg, 41% yield, m/z 387.0 [M+H]⁺ observed) as a white solid. The NMR for Compound 140 is described herein.

Example 16: Biological Example

Procedure for the Testing for Antinociception

The hot plate assay is adapted from the procedure originally described by O'Callaghan and Holtzman (JPET, 192, 497, 1975) and is commonly used to determine the potential analgesic efficacy of opioid agonists. The antinociceptive effect of the composition(s) described herein in the hot plate is expressed in % MPE (Maximum Possible Effect).

Rats (175-250 g) or mice (20-30 g) acclimated to the vivarium for at least 48 hr prior to behavioral testing. Test drugs were administered by the subcutaneous (SC) route. Animals were placed on the hot plate, which the temperature was set at 50-56° C., depending on the in vitro potency of the compound. A cutoff time of 30-60 seconds was used depending on the temperature of the hot plate so that the paws of the animal displaying analgesia, was not damaged by the heat stimulus. The cutoff time was considered a 100% response to the thermal insult. Prior to drug treatment, each animal was tested to determine the baseline response. Thirty minutes after drug administration, animals were re-tested. Dose response experiments were performed to evaluate the potency of the test compound when various doses were administered at the point when maximal analgesia is observed.

The % MPE was calculated according to the following formula: % MPE=[(Post drug latency-baseline latency)/(60 or 30-baseline latency)]×100

ED₅₀ values were calculated from the mean % MPE values for each group using log dose-response curves by least-squares regression analysis.

TABLE 4

COMPOUND	ED ₅₀ or % MPE
Morphine	3.8 mg/kg SC
Compound 81	100% at 10 mg/kg SC
Compound 122	1.1 mg/kg SC
Compound 28	1.2 mg/kg SC
Compound 145	5.9 mg/kg SC

Results are shown in Table 4. Naïve or control mice typically exhibit reaction times in the hot plate from 10-15 seconds. The ED₅₀ for morphine in the mouse hot plate was 3.8 mg/kg with full efficacy observed at a dose of 10 mg/kg

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SC. For comparison, Compound 122 and Compound 28 produced potent efficacy with an ED₅₀ of 1.1 and 1.2 mg/kg SC, respectively. These results demonstrate that Compound 122 and Compound 28 produced a more robust analgesic effect in the mouse hot plate assay compared to morphine.

Example 17: In Vivo Administration to Humans
(Prophetic Example)

One or more compounds will be administered in dosage range from 0.15 mg to 4 mg to human subject. The compound(s) will be administered as a continuous infusion over one hour. The dose may be escalated as deemed appropriate to obtain pain relief. Dose escalation will usually not exceed 5-fold as compared to the previous dose. Dosage amounts, however, may be repeated or decreased as deemed appropriate. The subjects will be tested for their ability to withstand or not appreciate pain as compared to a control (placebo) group.

The cold pain test has been shown to be a reproducible and sensitive measure of the effect of opiates and other centrally acting drugs (Van F and Rolan PE. The utility of the cold pain test to measure analgesia from intravenous morphine. *Br. J. Clin. Pharmacol.* 1996; 42: 663-664; Posner J. Pain Models in Healthy Volunteers. In: Nimmo W S, Tucker G, eds. *Clinical Measurement in Drug Evaluation*. 1991, Wolfe Publishing Limited, UK.; Wotherspoon H A, Kenny G N C, McArdle C S. Analgesic Efficacy of Controlled-Release DihydroCodeine. *Anaesthesia* 1991; 46: 915-917.; Lamb R J, Mercer A J, Posner J. The effect of lamotrigine (300 mg) and dipipanone (4 mg and 8 mg), alone and in combination, on the cold-pain test in healthy volunteers. *Br. J. Clin. Pharmacol.* 1994; 39: 539-588P.). In the test a subject's hand is immersed in cold water chilled to a range of 1 to 3° C. The initial sensation of cold is replaced by a deep burning discomfort in the hand which is mediated by nociceptors in veins. The discomfort gradually builds to a plateau over approximately 90 seconds and then either persists or decreases slightly. The stimulus is easily controlled and the response is reproducible. The technique has been shown to be sensitive to different doses of analgesic drugs.

During the cold pain test, the subject will sit down and place his/her non-dominant hand into a stirred, thermostatically controlled water bath at about 2° C. With the other hand the subject can adjust a visual analogue scale on a computer screen using the arrow keys on the keypad. The scale is labelled "no pain" at one end and "maximum pain" at the other end. The pointer will initially be at the "no pain" end and the subject will move the pointer across the line to rate their feelings continuously over the test period. At the end of 2 minutes the computer will automatically instruct the subject to remove his/her hand which can then be dried. The cold pain test has been used extensively in healthy volunteer studies and is non-invasive.

It is expected that the administration of the compound(s) will enable the human subject to feel no pain or less pain as compared to the control group.

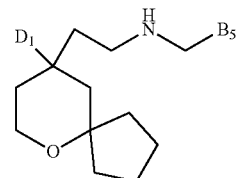
While the compounds described herein have been described with reference to examples, those skilled in the art recognize that various modifications may be made without departing from the spirit and scope thereof.

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All of the above U.S. patents, U.S. patent application publications, U.S. patent applications, foreign patents, foreign patent applications and non-patent publications referred to in this specification and/or listed in the Application Data Sheet are incorporated herein by reference, in their entirety.

What is claimed is:

1. A method of treating drug abuse comprising administering to a subject or a subject in need thereof a compound having the formula:



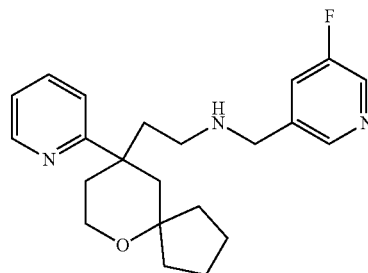
or a pharmaceutically acceptable salt thereof,

wherein:

D₁ is an optionally substituted aryl; and

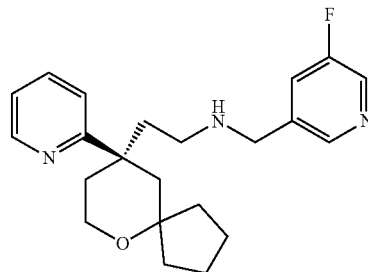
B₅ is an optionally substituted pyridyl.

2. The method of claim 1, wherein the compound has the formula



or a pharmaceutically acceptable salt thereof.

3. The method of claim 1, wherein the compound has the formula



or a pharmaceutically acceptable salt thereof.

4. A method of treating drug abuse comprising administering to a subject or a subject in need thereof a pharmaceutical composition comprising a compound having the formula: